Mangrove Loss & Blue Carbon Stock Analysis (1996–2020)

A Comparative Spatiotemporal Assessment of Mangrove Ecosystem Dynamics and Blue Carbon Implications in the Mekong Delta and the Mexican Pacific Coast

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Tools: R · QGIS · Global Mangrove Watch v3.0 · IPCC Tier 1

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1. Executive Summary

This report presents a spatiotemporal analysis of mangrove forest loss and associated blue carbon stock changes across two ecologically and economically significant regions: the **Mekong Delta (Vietnam)** and the **Mexican Pacific Coast (Oaxaca–Chiapas)**. Using satellite-derived data from the Global Mangrove Watch v3.0 (GMW; Bunting et al., 2022) and IPCC Tier 1 carbon coefficients, the analysis covers the period 1996–2020 across five time steps (1996, 2007, 2010, 2015, 2020).

Both regions experienced a continuous net decline in mangrove extent over the study period. The Mekong Delta lost **91.2 km^2** (–9.0%) of its mangrove cover, while the Mexican Pacific Coast lost **35.1 km^2** (–7.6%). These losses translated into estimated carbon stock reductions of **4.97 Mt C** and **1.71 Mt C**, respectively, representing a permanent reduction in ecosystem-level carbon sequestration capacity.

–91.2
km^2 (–9.0%)
Net Loss — Mekong Delta

–35.1
km^2 (–7.6%)
Net Loss — Mexican Pacific

4.97
Mt C
Carbon Lost — Mekong

1.71
Mt C
Carbon Lost — Mexic

2. Introduction

Mangrove forests are among the most carbon-dense ecosystems on Earth, storing up to five times more carbon per unit area than terrestrial tropical forests (Donato et al., 2011). This stored carbon (referred to as *blue carbon*) is sequestered in above-ground biomass and, critically, in deep organic-rich soils that can accumulate over millennia. When mangroves are cleared or degraded, this carbon is rapidly released back into the atmosphere, making mangrove loss a significant contributor to global greenhouse gas emissions.

Despite covering less than 0.5% of the world's ocean area, mangroves provide essential services including coastal protection, fisheries habitat, water filtration, and climate regulation. Global mangrove area has declined by approximately 20–35% over the past 50 years, driven primarily by aquaculture expansion, agricultural conversion, urban development, and unsustainable resource extraction (Hamilton & Casey, 2016).

This study focuses on two regions that represent contrasting socio-ecological contexts but share similar trajectories of mangrove loss:

Mekong Delta, Vietnam: One of Southeast Asia's largest mangrove systems, historically impacted by the shrimp aquaculture industry and hydrological alterations from upstream dams.

Mexican Pacific Coast (Oaxaca–Chiapas): A biodiversity hotspot containing significant mangrove patches under pressure from agricultural expansion and coastal development.

Key Finding: Mangrove loss is not merely a conservation concern; it is a measurable climate risk. Every square kilometer of mangrove lost releases an estimated 3,000–6,000 tonnes of stored carbon.

3. Data & Methods

3.1 Mangrove Extent Data

Mangrove extent and change data were obtained from the **Global Mangrove Watch v3.0** (GMW v3.0; Bunting et al., 2022), a satellite-derived global dataset produced using ALOS PALSAR L-band SAR imagery combined with optical data from Landsat. GMW v3.0 provides binary mangrove presence/absence maps at 25 m spatial resolution for 1996, 2007, 2010, 2015, and 2020.

Gain and loss pixels were identified through temporal comparison between consecutive epochs. Total mangrove area per region was computed in QGIS by summing all pixels classified as mangrove presence within each study boundary.

3.2 Blue Carbon Stock Estimation

Carbon stock was estimated using IPCC Tier 1 default values for two components:

• **Above-ground biomass carbon (AGB):** 159.4 t C/ha for tropical mangroves (IPCC, 2013)

• **Soil organic carbon (SOC):** 386 t C/ha (Hamilton & Friess, 2018)

Total carbon stock per epoch was calculated by multiplying the combined carbon density (545.4 t C/ha) by the mangrove extent for each year. Carbon lost was computed as the difference between the 1996 and 2020 carbon stock estimates.

3.3 Statistical Analysis

Trend analysis was performed using the non-parametric **Mann-Kendall test** (R package *Kendall*), appropriate for monotonic trend detection in small, non-normally distributed time series. The test evaluates whether the time series shows a statistically significant directional trend without assuming linearity.

Component	Source / Value	Reference
Mangrove extent	GMW v3.0, 25m SAR/optical	Bunting et al. (2022)
Above-ground C	159.4 t C/ha	IPCC (2013)
Soil organic C	386 t C/ha	Hamilton & Friess (2018)
Trend analysis	Mann-Kendall test (R)	Kendall package
GIS processing	QGIS 3.x + R (tidyverse, ggplot2)	Open source

Table 1. Summary of data sources and methods used in this analysis.

4. Study Areas

Figure M1. Mekong Delta Mangrove Dynamics 1996–2020. Cyan = gain; red = loss; dark green = stable mangrove. Data: Global Mangrove Watch v3.0. Basemap: Satellite imagery.

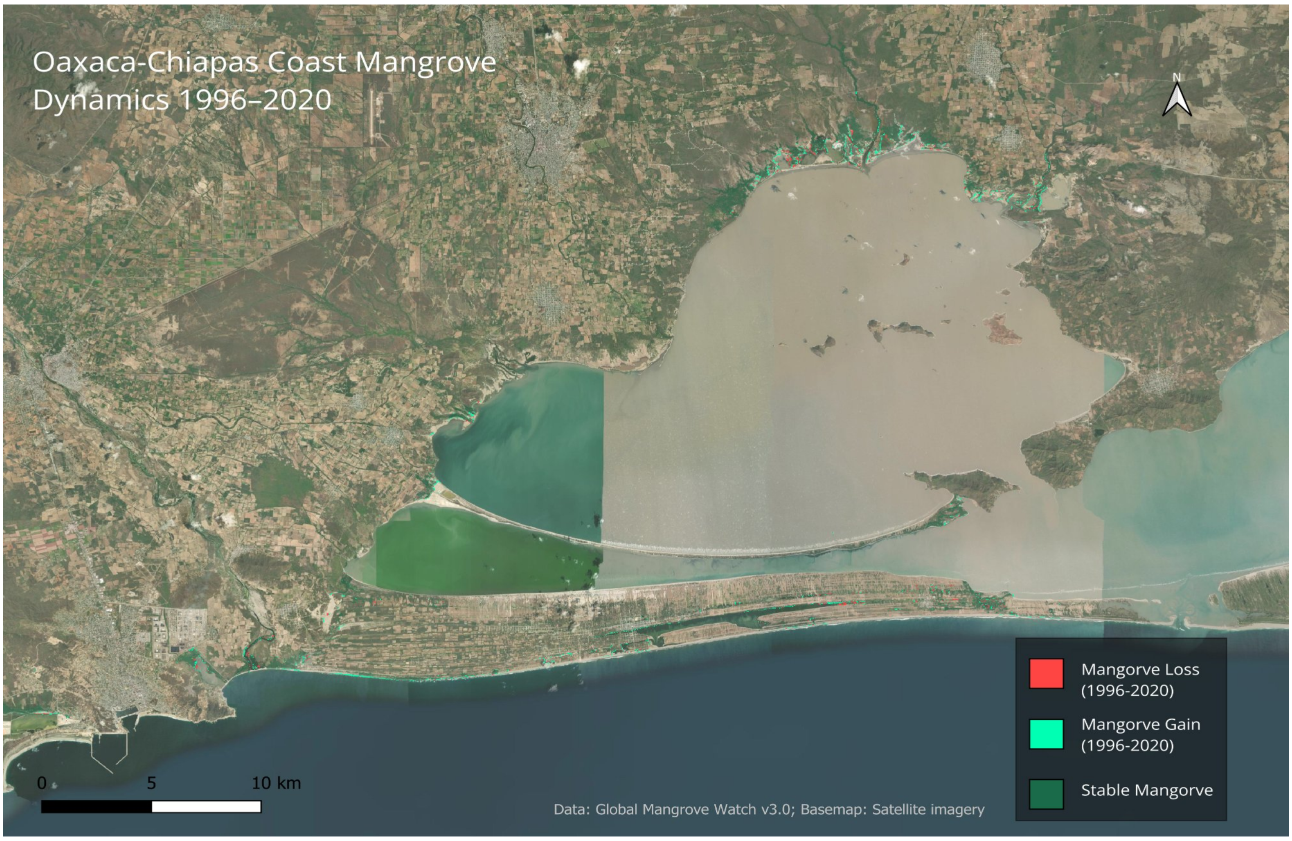


Figure M2. Oaxaca–Chiapas Coast Mangrove Dynamics 1996–2020. Cyan = gain; red = loss; dark green = stable mangrove. Data: Global Mangrove Watch v3.0. Basemap: Satellite imagery.

4.1 Mekong Delta, Vietnam

The Mekong Delta spans approximately 39,000 km^2 in southern Vietnam and is one of the most productive coastal ecosystems in Southeast Asia. Its mangrove forests are concentrated along the Ca Mau Peninsula and have been under intense pressure since the 1980s due to rapid expansion of intensive shrimp aquaculture. The delta's hydrology has also been altered by a cascade of upstream dams on the Mekong mainstem, reducing sediment supply critical for mangrove regeneration.

4.2 Mexican Pacific Coast (Oaxaca–Chiapas)

The Pacific coast of Oaxaca and Chiapas hosts a mosaic of coastal lagoons, estuaries, and mangrove patches that represent some of Mexico's most biodiverse coastal habitats. The region faces pressures from agricultural encroachment, tourism infrastructure development, and climate-driven disturbances including hurricanes and sea-level fluctuations. Mexico holds approximately 8% of the world's mangrove area, making its conservation a global priority.

5. Results

5.1 Mangrove Extent Trends (1996–2020)

Both regions showed consistent declines in mangrove extent across all time steps. The Mekong Delta began with 1,009.64 km^2 in 1996 and declined to 918.46 km^2 by 2020, representing a total loss of 91.2 km^2 at an average annual rate of –3.80 km^2/year. The Mexican Pacific Coast declined from 463.67 km^2 to 428.57 km^2 (–35.1 km^2; –1.46 km^2/year).

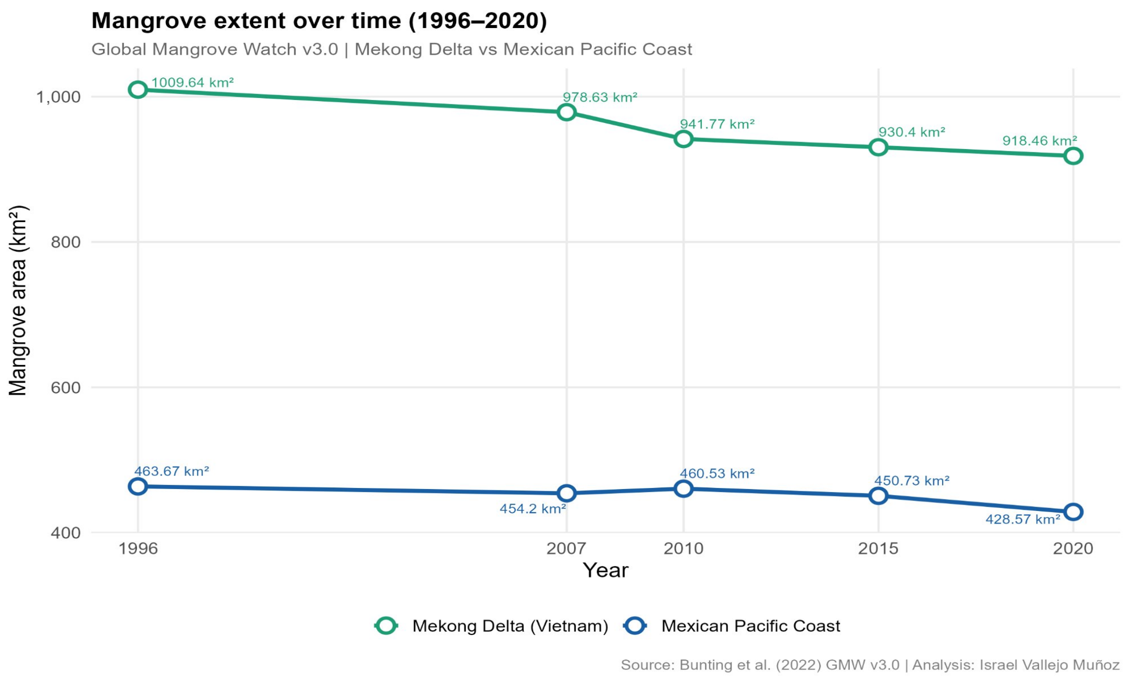


Figure 1. Mangrove extent over time (1996–2020) for the Mekong Delta (Vietnam) and Mexican Pacific Coast. Source: Global Mangrove Watch v3.0 (Bunting et al., 2022). Analysis: Israel Vallejo Muñoz.

5.2 Mann-Kendall Trend Analysis

The Mann-Kendall non-parametric trend test confirmed a statistically significant monotonic decline in the Mekong Delta ($z = –2.20$, $p = 0.027$, $\tau = –1.0$), indicating a perfect negative monotonic trend. The Mexican Pacific Coast also showed a negative trend ($z = –1.71$, $\tau = –0.80$), though at a marginally non-significant level ($p = 0.086$), consistent with a strong but slightly more variable decline.

Region	z	p-value	Tau (τ)	Interpretation
Mekong Delta	–2.20	0.027*	–1.00	Significant decline
Mexican Pacific	–1.71	0.086	–0.80	Strong negative trend

*Table 2. Mann-Kendall trend test results. *p < 0.05 denotes statistical significance.*

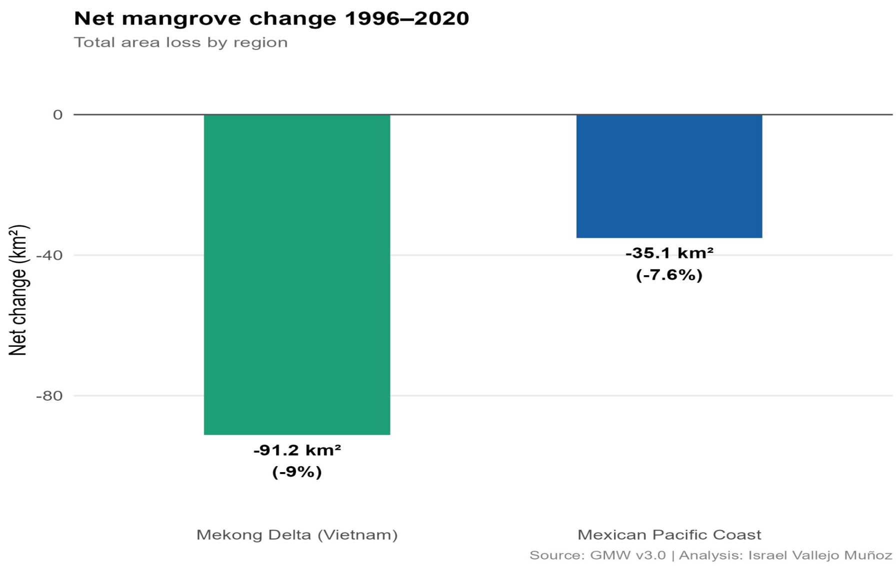


Figure 2. Net mangrove area change (km^2) between 1996 and 2020 by region. Source: GMW v3.0. Analysis: Israel Vallejo Muñoz.

5.3 Blue Carbon Stock Changes

The parallel decline in mangrove extent translated directly into reductions in blue carbon storage capacity. The Mekong Delta held an estimated 55.0 Mt C in 1996, declining to 50.1 Mt C by 2020, a net loss of **4.97 Mt C**. The Mexican Pacific Coast declined from 22.6 Mt C to 20.9 Mt C, losing **1.71 Mt C** over the same period.

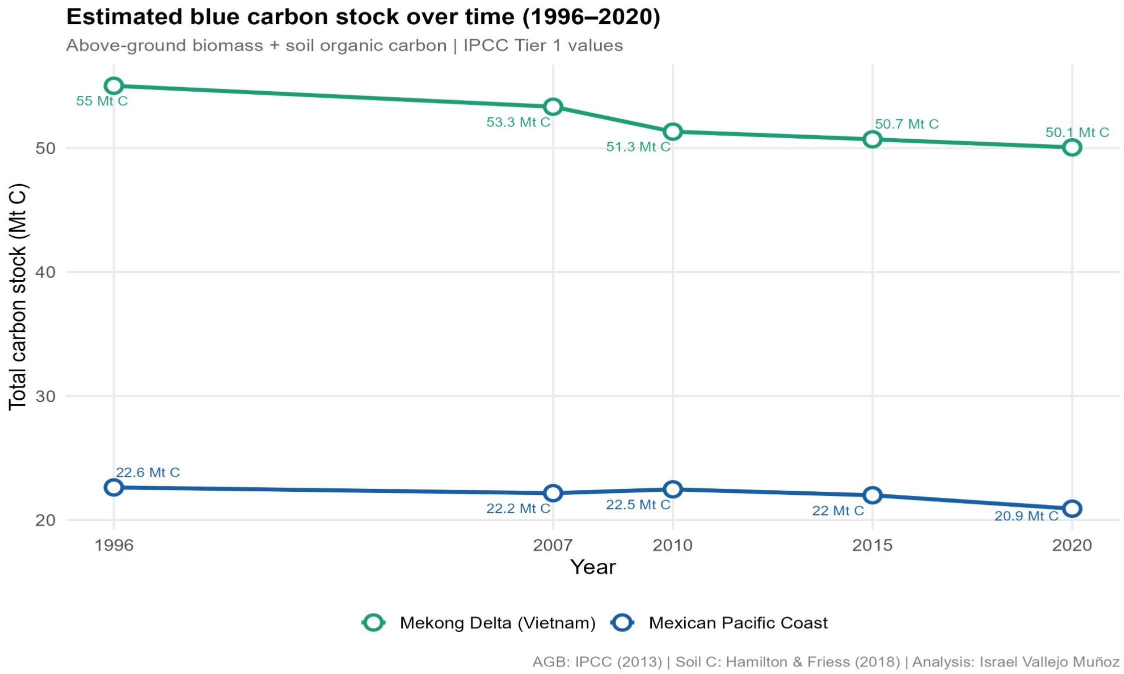


Figure 3. Estimated blue carbon stock (above-ground biomass + soil organic carbon) over time. AGB: IPCC (2013); Soil C: Hamilton & Friess (2018). Analysis: Israel Vallejo Muñoz.

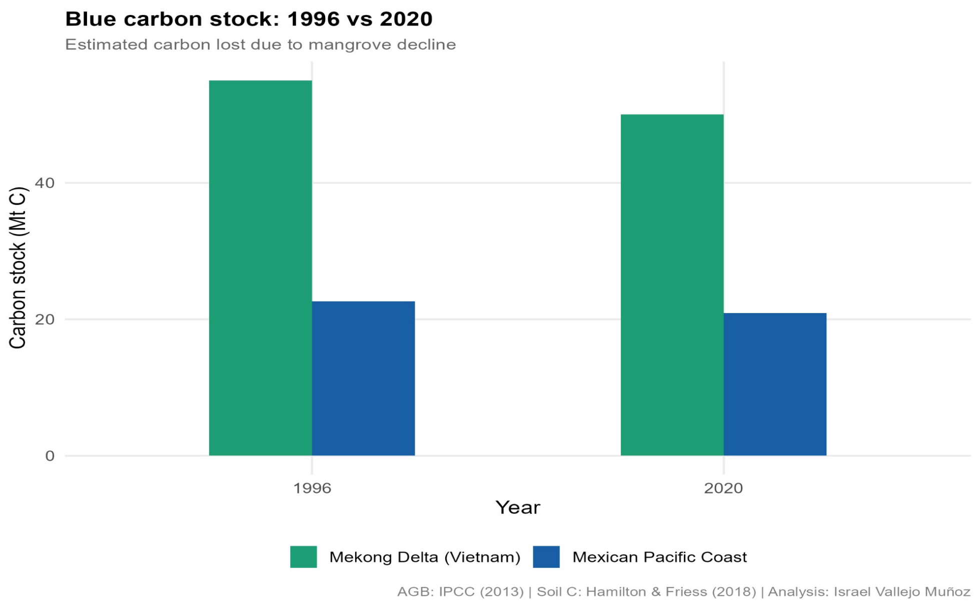
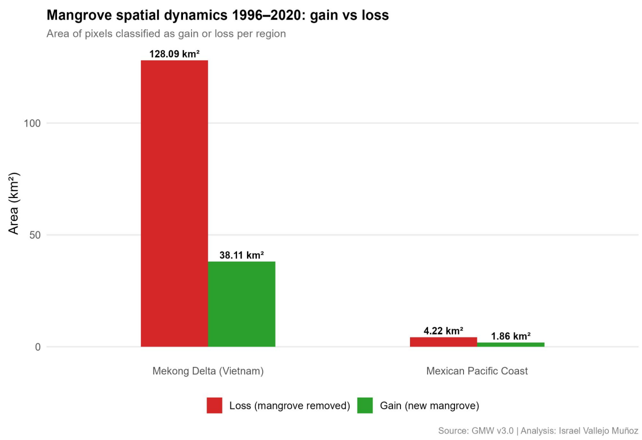


Figure 4. Blue carbon stock comparison between 1996 and 2020, illustrating total carbon lost due to mangrove decline. Analysis: Israel Vallejo Muñoz.

Key Finding: The Mekong Delta lost nearly 5 Mt of stored carbon over 24 years, equivalent to the annual CO■ emissions of approximately 3.6 million passenger vehicles.

5.4 Spatial Dynamics: Gain vs. Loss

Pixel-level gain/loss analysis revealed that in both regions, gross mangrove loss vastly exceeded gross gains. In the Mekong Delta, 128.09 km^2 were classified as loss versus only 38.11 km^2 as gain, with a loss-to-gain ratio of 3.4:1, with 77.1% of all changed pixels representing mangrove removal. In Mexico, the ratio was more moderate (4.22 km^2 loss vs. 1.86 km^2 gain), though loss still dominated at 69.4% of changed pixels.



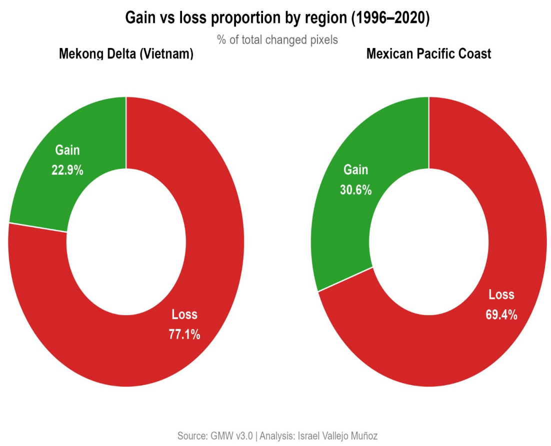


Figure 5 (left). Absolute gain and loss areas (km^2) by region. Figure 6 (right). Proportional gain vs. loss by region (% of total changed pixels). Source: GMW v3.0. Analysis: Israel Vallejo Muñoz.

6. Discussion

6.1 Drivers of Mangrove Loss

The contrasting rates of loss between the two regions reflect different underlying drivers. In the Mekong Delta, mangrove decline has been historically linked to large-scale aquaculture pond construction and hydrological changes, with the period 1996–2010 showing the steepest decline (–67.87 km^2 in 14 years). The partial stabilization observed after 2010 may reflect government replanting programs and stricter coastal protection policies implemented in Vietnam.

In Mexico, the loss trajectory is more gradual but persistent across all time steps, suggesting diffuse and ongoing pressures from agricultural expansion, tourism development, and climate-driven disturbances. The Mexican Pacific coast also showed a slightly higher relative gain proportion (30.6%) compared to the Mekong Delta (22.9%), which may indicate localized natural recovery or passive recolonization in disturbed areas.

6.2 Blue Carbon Implications

The estimated carbon losses underscore the climate relevance of mangrove conservation. Applying the global average above-ground + soil carbon density used here (~545 t C/ha), the combined loss of 126.3 km^2 across both regions represents approximately **6.68 Mt of carbon** no longer stored in living ecosystems. If released as CO_2, this would be equivalent to over 24.5 Mt CO_2e, highlighting the significant climate mitigation value of intact mangrove systems.

It should be noted that IPCC Tier 1 values represent regional averages and carry inherent uncertainty. Site-specific field measurements would be required for more precise carbon accounting, particularly for soil carbon stocks which can vary substantially with local conditions.

6.3 Methodological Considerations

The GMW v3.0 dataset provides a consistent, globally standardized baseline for mangrove monitoring. However, the 25m pixel resolution may underestimate gains and losses at patch edges, and the five available time steps limit the temporal resolution of the Mann-Kendall analysis (n = 5). Future work could integrate higher-resolution sensors (Sentinel-1/2, Planet) and include additional socioeconomic drivers to explain spatial patterns of change.

7. Conclusions

This analysis demonstrates that both the Mekong Delta and the Mexican Pacific Coast experienced significant, sustained mangrove loss between 1996 and 2020, with total reductions of 9.0% and 7.6% respectively. These losses carry quantifiable implications for blue carbon storage, coastal resilience, and biodiversity.

1. Both regions show statistically supported declining trends in mangrove extent (Mann-Kendall), with the Mekong Delta showing a perfectly monotonic decline ($\tau = -1.0$).

2. Combined blue carbon loss across both regions exceeds 6.68 Mt C, equivalent to over 24.5 Mt CO_2e if fully emitted.

3. Gross loss dominates over gross gain in both regions (Mekong 3.4:1; Mexico 2.3:1), indicating that natural recovery cannot offset ongoing anthropogenic pressures.

4. The Mekong Delta's post-2010 stabilization suggests that conservation policy can effectively slow degradation, offering a model for the Mexican Pacific context.

5. Satellite-derived data (GMW v3.0) combined with IPCC carbon accounting provides a reproducible, scalable framework for regional-scale mangrove carbon monitoring.

Key Finding: Protecting remaining mangrove forests in both regions is not only an ecological imperative but also a cost-effective climate mitigation strategy, given their exceptionally high carbon density.

References

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